ECE 329 Introduction to Electromagnetic Fields Spring 18

University of Illinois **He, Goddard, Ilie**

Exam 1

Wednesday, February 14, 2018 — 7:15-8:30 PM

Please clearly PRINT your name in CAPITAL LETTERS and circle your section in the boxes below.

Name:	Solutions
Section:	12 Noon 1 PM 2 PM

If you want your exam returned to a **different** section, write it here:

Special Return to Section:	12 Noon 1 PM 2 PM

This is a closed book exam and calculators/electronic devices are not allowed. You are allowed ONE double sided 3"x5" index card of notes. Please show all your work and make sure to include your reasoning for each answer. All answers should include **units** wherever appropriate. The exam is double sided. You may use the back of the exam as scratch paper.

Problem 1 (21 points)	
Problem 2 (25 points)	
Problem 3 (10 points)	
Problem 4 (14 points)	
Problem 5 (30 points)	
TOTAL (100 points)	

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1. (21 points) Multiple choice questions (3 points each). Circle only **ONE** letter next to the **ONE** correct statement. Note that most of these questions require some calculation to get the right answer, but you do not need to show your work to get credit. No partial credit is given on the multiple choice questions.

(i) Consider an inhomogeneous dielectric material (permittivity $\epsilon(x)$ varies with location). Let $\boldsymbol{E}$ be the electric field, V be the electrostatic potential. If the free charge density $\rho = 0$, which of the following equation is true?

(a) $\nabla V = 0$

(b) $\epsilon \nabla \cdot \boldsymbol{E} = 0$

(c) $\nabla^2 V = 0$

(d) $\nabla^2(\epsilon V) = 0$

(e) None of these

$0 = \rho = \nabla \cdot D = \nabla \cdot (\varepsilon E)$

$= \nabla \cdot (\varepsilon(-\nabla V)) = -\nabla\varepsilon \cdot \nabla V - \varepsilon \nabla^2 V$

(ii) For two perfect dielectrics with surface charge density $\rho_s = 0$ at the boundary, under electrostatic conditions, which of the following boundary condition statement is true ?

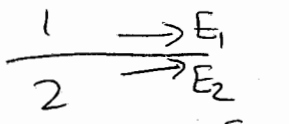
(a) The magnitude of $\boldsymbol{E}$ must always be continuous.

(b) The tangential component of $\boldsymbol{E}$ must always be continuous.

(c) The magnitude of $\boldsymbol{D}$ must always be continuous.

(d) The tangential component of $\boldsymbol{D}$ must always be continuous.

(e) None of these



$E_{1t} = E_{2t}$

(iii) For a dielectric media with permittivity $\epsilon = 5\epsilon_0$, under electrostatic condition, if the polarization field is $\boldsymbol{P}$, which of the following is the total electrical field $\boldsymbol{E}$ in the dielectric?

(a) $\frac{4\mathbf{P}}{\epsilon_0}$

(b) $\frac{2\mathbf{P}}{\epsilon_0}$

(c) $\frac{\mathbf{P}}{2\epsilon_0}$

(d) $\frac{\mathbf{P}}{4\epsilon_0}$

(e) None of these

$P = \varepsilon_0 \chi_e E$ $\quad D = \varepsilon E = \varepsilon_0 E + P$

$\Rightarrow E = \frac{P}{\varepsilon - \varepsilon_0} = \frac{P}{4\varepsilon_0}$

(iv) Consider the electric field $\mathbf{E} = < 2x + z^2,\ 3y^2,\ 2xz >$. Which statement is **FALSE**?

(a) The electric field is electrostatic in nature.

(b) The electric field admits a scalar potential $V(x, y, z)$.

(c) The charge density everywhere is zero.

(d) The path integral $\int_C \mathbf{E} \cdot d\mathbf{l} = 0$ for any curve C.

Note (a), (b) and (d) are equivalent definitions for a conservative field. So (c) must be it.

(d) is also correct. The original intent was for it to be the closed line integral, but due to a typesetting error, the circle was missing from the integral sign.

Check $\nabla \cdot D = 0$. Problem should say assume constant ε.

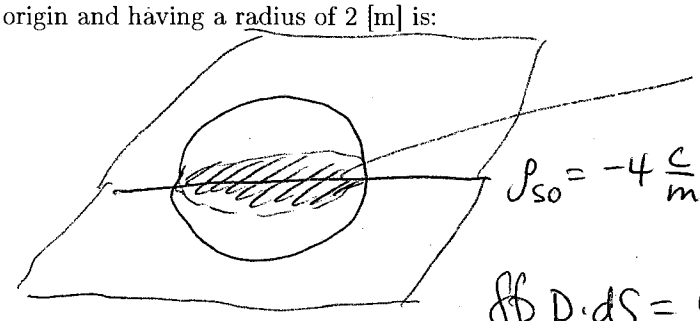
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Check $\nabla \cdot E = 0$.

$\nabla \cdot E = 2 + 6y + 2x \neq 0$ so $\rho \neq 0$

(v) The plane $z = 0$ contains a negative surface charge density $-4\ [\frac{C}{m^2}]$ and the rest of space is a dielectric with permittivity $\epsilon = 2\epsilon_0$. The outward electric flux $\int \mathbf{D} \cdot d\mathbf{S}$ across the closed spherical shell centered at the origin and having a radius of 2 [m] is:

(a) -64π [C]
(b) -16π [C]
(c) $-\frac{32}{3}\pi$ [C]
(d) -4π [C]
(e) $-\frac{8}{3}\pi$ [C]
(f) -1π [C]
(g) None of these



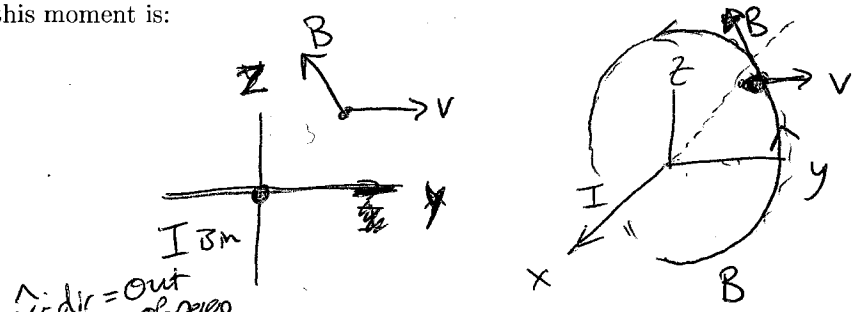
$Q_{enc} = \rho_{so} \cdot \pi r^2$
$= (-4)(\pi \cdot 4)$
$= -16\pi$

$\rho_{so} = -4\ \frac{C}{m^2}$

$\oint D \cdot dS = Q_{enc}$

(vi) A wire carries current $I = 1$[A] in the $+\hat{x}$ direction along the x-axis. A positive charge Q is moving with velocity $\mathbf{v} = 3\,\hat{y}[\frac{m}{s}]$ at the moment its position is $(x, y, z) = (0, 2, 2)$m. The force that the moving charge feels at this moment is:

(a) in the $+\hat{x}$ direction
(b) in the $-\hat{x}$ direction
(c) in the $+\hat{z}$ direction
(d) in the $-\hat{z}$ direction
(e) zero
(f) None of these



$F = qv \times B$ is in $+\hat{x}$ direction

(vii) Suppose you measure the components of magnetic flux density everywhere and observe that $B_x = x$ and $B_z = 0$. What can you conclude about B_y?

(a) $B_y = 0$
(b) $B_y = -1$
(c) $\frac{\partial B_y}{\partial y} = 0$
(d) $\frac{\partial B_y}{\partial y} = -1$
(e) $\nabla^2(B_y) = 0$
(f) None of these can be concluded.

$0 = \nabla \cdot B = \frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z}$ (0)

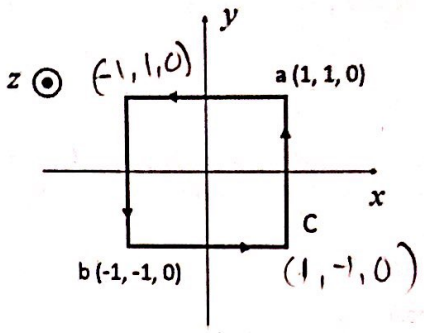
$= 1 + \frac{\partial B_y}{\partial y}$

$\Rightarrow \frac{\partial B_y}{\partial y} = -1$

(d) is the correct answer. The side calculation is correct but the wrong answer was circled.

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2. (25 points) Considering elelctric field $\mathbf{E}$ in free space and path C indicated in figure below, answer the following questions:



a) If the electric field is $\mathbf{E} = 2\hat{x} - 3\hat{y} + 5\hat{z}$ V/m,

i. (7 pts) what is the circulation $\oint \mathbf{E} \cdot d\mathbf{l}$ around path C? Show your work.

Approach 1 $\oint \vec{E} \cdot d\vec{l} = \int_{-1}^{1} 2\,dx + \int_{-1}^{1} (-3)\,dy + \int_{1}^{-1} 2\,dx + \int_{1}^{-1} (-3)\,dy$

$= 4 - 6 - 4 + 6$

$= 0$

Approach 2: Also, $\nabla \times \vec{E} = 0$

$\therefore \oint \vec{E} \cdot d\vec{l} = 0.$

Your Answer (include appropriate units): $\oint \mathbf{E} \cdot d\mathbf{l} =$ 0

ii. (7 pts) what is the electrostatic potential drop from point a to b (i.e. $V(a) - V(b)$)? Show your work.

$V(a) - V(b) = -\int_{b}^{a} \vec{E} \cdot d\vec{l}$

$= -\int_{-1}^{1} 2\,dx - \int_{-1}^{1} (-3)\,dy$

$= -4 + 6$

$= +2V$

Your Answer (include appropriate units): $V(a) - V(b) =$ 2 V

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b) If instead, the curl of the elelctric field is $\nabla \times \mathbf{E} = 2\hat{x} - 3\hat{y} + 5\hat{z}$ V/m^2,

i. (8 pts) what is the circulation $\oint \mathbf{E} \cdot d\boldsymbol{l}$ around path C? Show your work.

By Stokes's theorem,

$$\oint \vec{E} \cdot d\vec{l} = \iint_S (\nabla \times \vec{E}) \cdot d\vec{S}$$

$$= \iint (2\hat{x} - 3\hat{y} + 5\hat{z}) \cdot (dx\,dy\,\hat{z})$$

$$= \int_{-1}^{1} \int_{-1}^{1} 5\,dx\,dy$$

$$= 5 \times 4$$

$$= 20\,V$$

Your Answer (include appropriate units): $\oint \mathbf{E} \cdot d\boldsymbol{l}$ = 20 V

ii. (3 pts) which of the following statements is correct about electrostatic potential drop from point a to b (i.e. $V(a) - V(b)$)?

Non conservative field.

A. $V(a) - V(b) = 2$V.

B. $V(a) - V(b) = -2$V.

C. $V(a) - V(b) = 10$V.

D. $V(a) - V(b) = -10$V.

(E) $V(a) - V(b)$ is undefined.

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3. (10 points) Compute the total outward electric flux from the cube $0 < x, y, z < 1$ m, containing the volumetric charge density of $\rho = 16yz^3 \left[\frac{\mu C}{m^3}\right]$.

$$\Psi_e = \oiint D \cdot dS = Q_{enc}$$

$$Q_{enc} = \iiint \rho \, dV$$

$$= \int_0^1 \int_0^1 \int_0^1 16 y z^3 \, dx \, dy \, dz$$

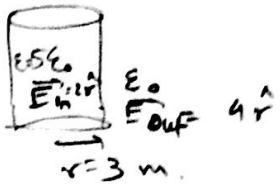
$$= 16 \times \left| \frac{y^2}{2} \right|_0^1 \left| \frac{z^4}{4} \right|_0^1 = 2$$

Your Answer (include appropriate units): $\int \mathbf{D} \cdot d\mathbf{S} =$ 2 μC

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4. (14 points) An electric field is observed to point radially outwards in **cylindrical** coordinates everywhere. The region $r < 3$ m is filled with a dielectric with permittivity $\epsilon = 5\epsilon_0$ while the region $r > 3$ m consists of free space ($\epsilon = \epsilon_0$). If the electric field is $\mathbf{E} = 4\hat{r}$ $[\frac{V}{m}]$ just outside the cylindrical shell $r = 3$ m and is $\mathbf{E} = 2\hat{r}$ $[\frac{V}{m}]$ just inside this shell, find the surface charge density ρ_s on the cylindrical shell.

Using the normal boundary condition,



$$\hat{a}_n \cdot (\vec{D}_{out} - \vec{D}_{in}) = \rho_s$$

$$\hat{r} \cdot (4\epsilon_0 \hat{r} - 2\epsilon_r \hat{r}) = \rho_s$$

$$\hat{r} \cdot (4\epsilon_0 \hat{r} - 10\epsilon_0 \hat{r}) = \rho_s$$

$$-6\epsilon_0 = \rho_s$$

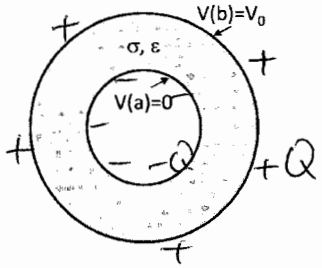
$$\therefore \boxed{\rho_s = -6\epsilon_0 \ C/m^2}$$

Your Answer (include appropriate units): $\rho_s =$	$-6\epsilon_0 \ C/m^2$.

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5. (30 points) A perfectly conducting **spherical** shell of radius b is held at a constant potential $V_0 > 0$ while a concentric perfectly conducting **spherical** shell of radius a (where $a < b$) is grounded. The region in between the two shells $a < r < b$ contains a material with conductance $\sigma > 0$ and permittivity $\epsilon = \epsilon_0$. You make measurements of the electrical properties of this configuration and find that there is a total charge of $Q > 0$ on one shell and $-Q < 0$ on the other and moreover that the electric field $\mathbf{E}$ and the volumetric charge density ρ are zero for $r < a$ and $r > b$.



a) (2 pts) (circle one) TRUE or (FALSE) The inner shell is the one with the positive charge.

b) (2 pts) (circle one) TRUE or (FALSE) The reason that the electric field for the region $r < a$ is zero is that the inner shell is grounded.

c) (4 pts) Write the electric field vector $\mathbf{E}$ in the region $a < r < b$ in terms of Q and other parameters.

Your Answer (include direction and appropriate units): $\mathbf{E} =$ $\frac{-Q}{4\pi\varepsilon r^2}\hat{r}$ [V/m]

d) (7 pts) Using Ohm's Law ($\mathbf{J} = \sigma\mathbf{E}$) and the relation between current I and current density $\mathbf{J}$, calculate the total current I flowing from the inner shell to outer shell that crosses a spherical shell of radius r_0 where $a < r_0 < b$. (**Hint:** because the lines of current flow are continuous, your answer will not depend on r_0).

$J = \frac{-\sigma Q}{4\pi\varepsilon r^2}\hat{r}$ (flows inward), It is constant on a sphere of radius r_0.

dS points outward

$$I = \iint J \cdot dS = \iint |J||dS| \cdot (-1) = -|J| \iint dS = -|J| \cdot \text{Area of sphere}$$

$$= -\left|\frac{-\sigma Q}{4\pi\varepsilon r_0^2}\right| \cdot 4\pi r_0^2 = -\frac{\sigma Q}{\varepsilon} \quad \text{(current flows inward)}$$

Your Answer (include appropriate units): $I =$ $-\frac{\sigma Q}{\varepsilon}$ [A]

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e) (13 pts) Use the electric field from part (c) and the definition of the electrostatic potential V to calculate the voltage difference $V(b) - V(a)$ in terms of Q and other parameters. Next, solve for Q in terms of V_0 by using the fact that $V_0 = V(b) - V(a)$.

$$V(b) - V(a) = -\int_a^b \vec{E} \cdot d\vec{\ell} \qquad d\vec{\ell} = dr\,\hat{r}$$

$$= -\int_a^b \left(\frac{-Q}{4\pi\varepsilon r^2}\hat{r}\right) \cdot dr\,\hat{r}$$

$$= +\int_a^b \frac{Q}{4\pi\varepsilon}\frac{1}{r^2}dr = \frac{Q}{4\pi\varepsilon}\left(\frac{-1}{r}\right)\Big|_a^b$$

$$= \frac{Q}{4\pi\varepsilon}\left(\frac{1}{a} - \frac{1}{b}\right)$$

$$V_0 = \frac{Q}{4\pi\varepsilon}\left(\frac{b-a}{ab}\right)$$

$$\Rightarrow Q = 4\pi\varepsilon V_0\left(\frac{ab}{b-a}\right)$$

Your Answer (include appropriate units): $V(b) - V(a) =$ $\frac{Q}{4\pi\varepsilon}\left(\frac{b-a}{ab}\right)$ [V]
Your Answer (include appropriate units): $Q =$ $4\pi\varepsilon V_0\left(\frac{ab}{b-a}\right)$ [C]

f) (2 pts) Compute the conductance G for this configuration.

$$G = \frac{I}{V_0} = \frac{+\sigma Q}{\varepsilon V_0} = \frac{+\sigma}{\varepsilon}\left(4\pi\varepsilon\frac{ab}{b-a}\right)$$

Your Answer (include appropriate units): $G =$ $\sigma \cdot 4\pi\frac{ab}{b-a}$ [Siemens]

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